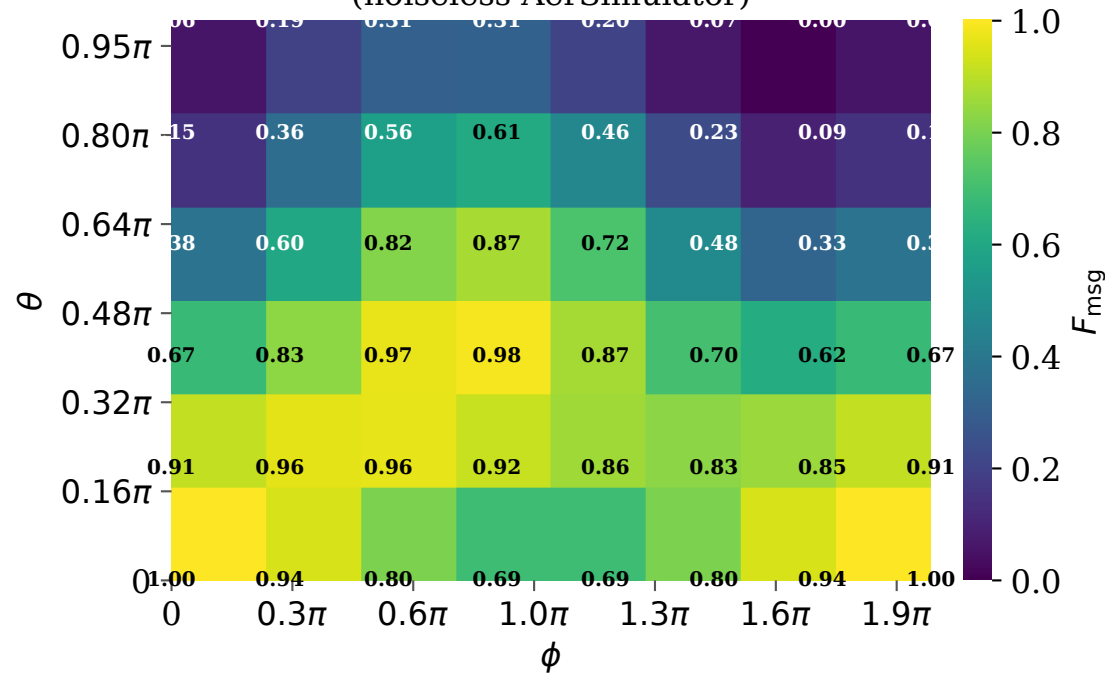


Hardware vs Simulator: $F_{\text{msg}}(\theta, \phi)$ parameter sweep
 ibm_torino, 6×8 grid, Spearman $r = 0.950$ ($p=7.92\text{e-}25$)

(a) Simulator $F_{\text{msg}}(\theta, \phi)$
 (noiseless AerSimulator)



(b) Hardware $F_{\text{msg}}(\theta, \phi)$
 (ibm_torino, 2000 shots/basis/point, DD=XX)

